

Uptake of Waterborne Methylmercury by Rainbow Trout (*Salmo gairdneri*) in Relation to Oxygen Consumption and Methylmercury Concentration

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Oxygen consumption and uptake of waterborne methylmercury were measured for rainbow trout (*Salmo gairdneri*) forced to swim at sustained swimming speeds at 10 and 20°C. The concentrations of methylmercury used ($< 8 \mu\text{g Hg} \cdot \text{L}^{-1}$) did not affect oxygen consumption. The log of mass specific oxygen consumption increased linearly with relative swimming speed. Standard metabolic rates were significantly different ($P < 0.05$) at 10 and 20°C (111 and 192 $\text{mg O}_2 \cdot \text{kg}^{-1} \cdot \text{h}^{-1}$, respectively) but the rate of increase in oxygen consumption with swimming speed was not significantly different between temperatures. The rate of methylmercury uptake was positively correlated with both oxygen consumption and methylmercury concentration. Multiple linear regression equations relating the logs of rate of methylmercury uptake, rate of oxygen consumption, and methylmercury concentration at 10 and 20°C were contiguous and a single equation described the relationship at both temperatures. The efficiency of methylmercury uptake relative to oxygen was ~ 0.25 at both temperatures. Assuming an oxygen percent utilization of 33%, the percent utilization of methylmercury from water was $\sim 8\%$.

Key words: methylmercury, uptake, respiration, oxygen consumption, rainbow trout, bioaccumulation, pollutant

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Chez des truites arc-en-ciel (*Salmo gairdneri*) forcées à une nage soutenue à 10 et 20°C, nous avons mesuré la consommation d'oxygène et l'absorption de mercure-méthyl. Les concentrations de mercure-méthyl utilisées ($< 8 \mu\text{g Hg} \cdot \text{L}^{-1}$) n'influencent pas la consommation d'oxygène. Le log de la consommation d'oxygène par rapport à la masse augmente en fonction linéaire de la vitesse de nage relative. Les taux métaboliques normaux sont significativement ($P < 0,05$) différents à 10 et 20°C (111 et 192 $\text{mg O}_2 \cdot \text{kg}^{-1} \cdot \text{h}^{-1}$, respectivement), mais le taux d'augmentation de la consommation d'oxygène en fonction de la

vitesse de nage ne diffère pas de façon significative à ces deux températures. Il y a corrélation positive entre le taux d'absorption de mercure-méthyl, d'une part, et la consommation d'oxygène et la concentration de mercure-méthyl, d'autre part. Les équations de régressions linéaires multiples reliant les logs du taux d'absorption du mercure-méthyl, les taux de consommation d'oxygène et la concentration de mercure-méthyl à 10 et 20°C sont contiguës, et une seule équation décrit la relation à l'une et l'autre température. L'efficacité d'absorption du mercure-méthyl en fonction de l'oxygène est ~ 0.25 aux deux températures. Si l'on suppose une utilisation de l'oxygène de 33%, le pourcentage d'utilisation du mercure-méthyl de l'eau est environ 8%.

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THE gills of teleost fishes are central to the exchange of dissolved gases, ions, and organic molecules (Conte 1969; Foster and Goldstein 1969; Randall 1970). The mechanisms of exchange may range from passive, including exchange and facilitated diffusion, to active carrier-mediated transport. Passive fluxes across the gills are governed by chemical and electrical gradients and by the physical properties of the substance exchanged (Hunn and Allen 1974). In particular, lipid soluble molecules may be absorbed selectively from solution by surface absorption and partitioning to the lipoprotein cell membrane (Bryan 1979). Both active and passive exchange are affected by the volume of water passed over the gills, which is determined by metabolic rate (Jones and Randall 1978).

Lipophilic and amphiphilic molecules once across the gills are transported by the circulatory system to other tissues (Olsen et al. 1973). The distribution of these compounds reflects their affinity to specific tissues and the pattern of circulation, while the concentration of the compound attained by the tissue depends on both distribution and the rate of metabolism or elimination from the tissue (deFreitas and Norstrom 1974). For compounds that are only slowly metabolized or eliminated from the fish, tissue concentrations substantially above ambient may be accumulated. The concentrations of many xenobiotics such as DDT, PCBs, and methylmercury attained by fish in contaminated waters (Hamelink et al. 1977; Norstrom et al. 1976) are examples of this accumulation process.

Numerous studies have demonstrated that fish readily accumulate waterborne xenobiotics which themselves may interfere with respiration (Hughes 1973). However, relatively few have attempted to relate uptake of these compounds to respiratory rate. The potentiation of the toxicity of a number of compounds to rainbow trout by low dissolved oxygen concentration was attributed to a coincident increase in respiratory volumes (Lloyd 1961). Both oxygen consumption and uptake of waterborne DDT increased with body weight to a similar exponent in mosquito fish, *Gambusia affinis* (Murphy and Murphy 1971). More recently, the uptake of waterborne xenobiotics has been related directly to oxygen consumption in restrained salmonids (Boddington et al. 1979; McKim 1981).

Our experiments were designed to test the relationships among oxygen consumption, methylmercury concentration, and uptake in rainbow trout, *Salmo gairdneri*, over a broad range of metabolic rates achieved by forcing the fish to swim at speeds between low-sustained and near-critical velocities at two temperatures.

Materials and Methods

Rainbow trout (4–10 g) were purchased from a private hatchery (Goosens, Otterville, Ont.), tagged (Floy fingerling tags – Floy Tag Manufacturing Ltd., Seattle, WA), and held in 50-L circular troughs supplied with aerated well water ($11 \pm 1.0^\circ\text{C}$). A submersible pump installed in each trough created a current of ~ 8 cm/s. Fish were acclimated to either 10 or 20°C by adjusting temperature at the rate of $1^\circ\text{C}/\text{d}$, then maintained at that temperature ($\pm 0.5^\circ\text{C}$) for at least 2 wk prior to an experiment. Conductivity, total hardness, and pH of the well water ranged from 0.64 to 0.72 mS/cm, 375 to 390 mg/L, and 8.0 to 8.2, respectively (APHA et al. 1971).

Fish were fed a commercial trout diet (GRT-78G – Martins Feed Mills, Elora, Ont.) once every 2nd day at a meal size equivalent to 2% wet body weight. Photoperiod was maintained at 16 h of light with a gradual change of intensity for $\frac{1}{2}$ h at each end of the light regime.

Individual fish were transferred from a holding tank to a modified Blazka-type respirometer (Waiwood and Beamish 1978) at $\sim 18:00$ the day before an experiment. Aerated well water from a temperature-regulated head tank flowed to each respirometer at a rate of ~ 150 mL/min. The respirometers were submerged in a water bath, with both bath and head tank maintained at the acclimation temperature of the fish. Fish were required to swim at ~ 10 cm $\cdot$ s $^{-1}$ (1.2 body lengths (BL) $\cdot$ s $^{-1}$) overnight. At 09:00 the next day, the water velocity was increased over a 5-min period to either 20, 30, or 40 cm $\cdot$ s $^{-1}$ and maintained at that speed throughout the experiment. These velocities were selected to provide a broad range of aerobic metabolic rates with the highest velocity equivalent to $\sim 80\%$ of critical speed (Waiwood and Beamish 1978). One hour after speed adjustment, dissolved oxygen concentration of the outflow of water of the respirometer was determined and the inflow stopped for an interval of 1 or 2 h at 20 and 10°C , respectively. Dissolved oxygen concentration was measured with an oxygen electrode and analyser (YSI, model 54) calibrated daily with the sodium azide modification of the Winkler method (APHA et al. 1971). The maximal cross-sectional area of the trout as determined from the relationship of cross-sectional area and weight of fish of the same stock (Waiwood 1977) was less than 10% of the cross-sectional area of the swimming chamber. As speed corrections should be small enough to ignore for fish occupying less than 10% of the area of the swimming chamber (Webb 1975), measured water velocities were used without correction.

At the end of the interval, water flow through the respirom-

eter was resumed and the dissolved oxygen concentration of the outflow determined. Throughout all tests, dissolved oxygen concentrations remained above 70% of air saturation. The difference in oxygen content between the beginning and end of the test interval, minus the aerobic demands of the chamber water in the absence of fish, provided the measurement of oxygen consumption.

Oxygen consumption of each fish was determined for three intervals. Between determinations, the flow of aerated water through the chamber was resumed for 1 h to allow dissolved oxygen to return to at least 90% of air saturation. Immediately preceding the third interval, [^{203}Hg]methylmercury (0.5–2.5 mL) was injected into the inflow water 1 min before the system was closed. Duplicate water samples (5 mL) were collected before injection, at 0 and 10 min and at the end of the time interval for determination of methylmercury concentration. At the end of the third interval, the fish was immediately removed from the respirometer and transferred to a holding tank. Within 24 h of the experiment, the fish was weighed and measured to the nearest 0.1 g and mm, respectively, and the uptake of methylmercury determined by whole body gamma counting. As the biological half-life of methylmercury in fish of this size is greater than 50 d (deFreitas et al. 1975), a correction for elimination before counting was not required.

Methylmercuric chloride labeled with ^{203}Hg (specific activity 17.5–87.5 MBq/mg Hg) (New England Nuclear, Montreal, P.Q.) was maintained as an aqueous solution at a concentration of $\sim 0.4 \text{ MBq} \cdot \text{mL}^{-1}$. At this concentration the level of ^{203}Hg -labeled impurities remained at less than 4% as measured by extraction with acidified benzene (Westoo 1973).

Methylmercury in water and fish was measured by gamma spectrometry of the [^{203}Hg]methylmercury. Water samples were counted by an automated gamma counter with a 7.5-cm-diameter, 7.5-cm-high, well-shaped NaI(Tl) crystal (Nuclear Chicago model 4233). Fish were counted in a manual whole body counter with two $5 \times 5 \times 15 \text{ cm}$ NaI(Tl) crystals as the detection unit. The rectangular faces of the crystals were horizontally opposed with a 5-cm space between crystals. The detection unit was shielded with lead bricks and the output of the two crystals summed and connected to a preamplifier, amplifier, single channel analyser, scaler, and timer (Harshaw detectors and electronics).

Fish were centered in a cylindrical acrylic container filled with uncontaminated water. The container was closed and placed in a fixed groove with the fish midway between the NaI(Tl) crystals. Three 1-min counts were made of each fish: fish counted at greater than 3000 cpm (background less than 250 cpm) and replicate counts varied by less than 3%. In a preliminary study, counting efficiency was not affected by fish size for fish from less than 5 to 50 g. Phantoms containing known quantities of nuclide were prepared from each batch of isotope and used as internal standards to correct for radioactive decay and interrelate counts on the deep well and whole body counters. Results, corrected for background, counting efficiency, and radioactive decay are expressed as nanograms of methylmercury.

The concentration of [^{203}Hg]methylmercury in a closed respirometer, without fish, decreased rapidly (Fig. 1). This loss was apparently due to adsorption and absorption to the acrylic

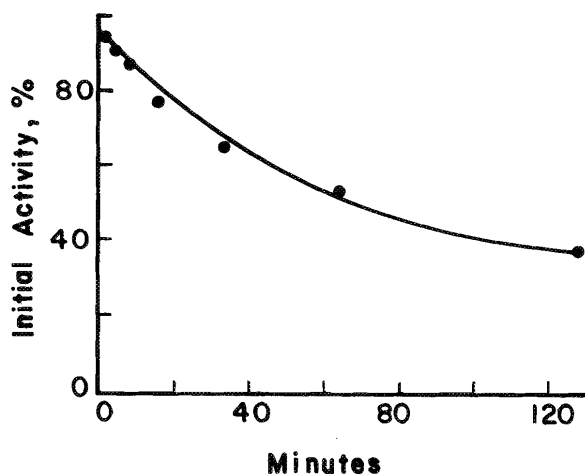


FIG. 1. Decrease in concentration of [^{203}Hg]methylmercury in water from a closed respirometer during a 128-min period (mean of 14 trials).

walls of the respirometer. An exponential decay function adequately described this loss with correlation coefficients of 0.96 or greater obtained in individual linear regressions of log concentration of isotope and time (14 trials, samples taken as indicated in Fig. 1). It was assumed that a similar exponential decay function would apply to the decrease in nuclide concentration in experiments with a fish in the respirometer as methylmercury uptake by the fish constituted less than 3% of the quantity of isotope introduced. The concentration of methylmercury experienced by the fish was thus determined as the mean of the logarithm of concentrations of [^{203}Hg]methylmercury at the beginning and end of exposure. Initial concentrations of methylmercury were in the range of $1\text{--}10 \mu\text{g CH}_3\text{Hg} \cdot \text{L}^{-1}$ with average concentrations from 0.5 to $4.0 \mu\text{g CH}_3\text{Hg} \cdot \text{L}^{-1}$. Water samples taken before injection of nuclide were counted at background rates as did fish held in the respirometers but not exposed to isotope.

The efficiency of methylmercury uptake relative to oxygen ($E(p/o)$) was estimated as:

$$E(p/o) = (\Delta P/C_{pw})/(Q/[O_2])$$

where ΔP is the rate of methylmercury uptake of the fish ($\text{ng CH}_3\text{Hg} \cdot \text{h}^{-1}$), C_{pw} is the methylmercury concentration ($\text{ng CH}_3\text{Hg} \cdot \text{mL}^{-1}$), Q is the rate of oxygen consumption of the fish ($\text{mg O}_2 \cdot \text{h}^{-1}$) and $[O_2]$ is the mean concentration of oxygen ($\text{mg O}_2 \cdot \text{mL}^{-1}$).

Comparisons between means were made using appropriate T -tests at $P < 0.05$. Least squares regressions were compared using either large sample Z tests or dummy variables (Kleinbaum and Kupper 1978). All regressions were significant at $P < 0.05$. The log transformation was used as indicated by residual plots.

Results

Oxygen consumption of rainbow trout was not affected by methylmercury at the concentrations and durations of exposure used. At both 10 and 20°C, the rate of oxygen con-

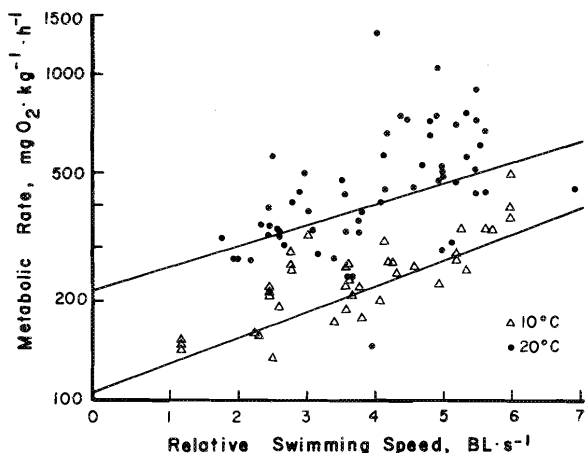


FIG. 2. Increase in mass specific oxygen consumption with a relative swimming speed of ~6.5-g rainbow trout at 10 and 20°C.

sumption for the interval in which methylmercury was added was not significantly different from the preceding interval ($P < 0.05$).

The mass specific rate of oxygen consumption increased with relative swimming speed at both 10 and 20°C with values obtained for individual fish shown in Fig. 2. The relationships are described by the equations:

$$(1) \quad 10^{\circ}\text{C} \quad \log M = 2.030 + 0.082 S \\ (R^2 = 0.60; n = 38)$$

$$(2) \quad 20^{\circ}\text{C} \quad \log M = 2.330 + 0.068 S \\ (R^2 = 0.25; n = 60)$$

where M is the mass specific rate of oxygen consumption ($\text{mg O}_2 \cdot \text{kg}^{-1} \cdot \text{h}^{-1}$) and S is the relative swimming speed ($\text{BL} \cdot \text{s}^{-1}$). Fish swam well at all speeds with no obvious differences in swimming behavior between temperatures. The mean size of fish was similar at both temperatures although fish at 20°C were somewhat more variable (6.7 ± 0.28 g (mean $\pm$ SE) and 76 ± 1.0 mm at 10°C; 6.3 ± 0.36 g and 78 ± 1.3 mm at 20°C).

The increased variability of metabolic rate of fish at 20°C is also apparent in the lower R^2 of equation (2). The rate of increase in oxygen consumption with swimming speed was not significantly different ($P < 0.05$) at 10 and 20°C. However, standard metabolic rates, calculated by extrapolation of the linear regressions to zero swimming speed (Brett 1965), were significantly different ($P < 0.05$) at 10 and 20°C (107 and 214 $\text{mg O}_2 \cdot \text{kg}^{-1} \cdot \text{h}^{-1}$ at 10 and 20°C, respectively), thus equations (1) and (2) were distinct. At swimming speeds of 6 $\text{BL} \cdot \text{s}^{-1}$, approximately critical speed for fish of this size (Waiwood and Beamish 1978), calculated metabolic rates are 332 and 545 $\text{mg O}_2 \cdot \text{kg}^{-1} \cdot \text{h}^{-1}$ at 10° and 20°C, respectively. The increase from standard to active metabolic rates or metabolic scope (Fry 1947) of the fish was 225 $\text{mg O}_2 \cdot \text{kg}^{-1} \cdot \text{h}^{-1}$ at 10° and 331 $\text{mg O}_2 \cdot \text{kg}^{-1} \cdot \text{h}^{-1}$ at 20°C.

The rate of methylmercury uptake was positively related to both rate of oxygen consumption and methylmercury concen-

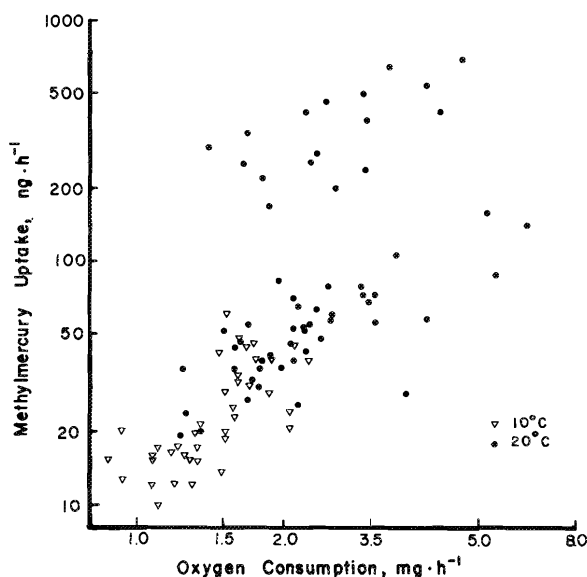


FIG. 3. Effect of oxygen consumption on the rate of methylmercury uptake of rainbow trout at 10 and 20°C.

tration as seen in Fig. 3 and 4, respectively. The multiple linear regression equations describing the relationship among these variables at each temperature are:

$$(3) \quad 10^{\circ}\text{C} \quad \log \Delta P = 1.326 + 1.147 \log Q \\ + 0.724 \log Cpw \\ (\text{multiple } R^2 = 0.74; n = 38)$$

$$(4) \quad 20^{\circ}\text{C} \quad \log \Delta P = 1.479 + 1.106 \log Q \\ + 1.197 \log Cpw \\ (\text{multiple } R^2 = 0.88; n = 60)$$

where ΔP , Q , and Cpw are as defined earlier.

The regression equations at each temperature are contiguous and can be reduced to a common regression which is not significantly different from the individual regressions ($P < 0.05$), yet had a higher R^2 than regressions at each temperature. The regression equation describing the pooled data is:

$$(5) \quad \log \Delta P = 1.457 + 1.097 \log Q + 1.152 \log Cpw \\ (\text{multiple } R^2 = 0.91; n = 98).$$

In equations (3), (4), and (5), both rate of oxygen consumption and methylmercury concentrations are highly significant factors ($P < 0.001$) in the prediction of rate of methylmercury uptake. Further, the dependent variables are largely independent in their contribution to the regression, with the variability (R^2) of regressions using either variable by itself approximately one-half the multiple R^2 using both variables.

Equation (5) could be further simplified by replacing the terms $\log Q$ and $\log Cpw$ with a combined term $\log (Q \times Cpw)$ as the coefficients of the above terms were not signifi-

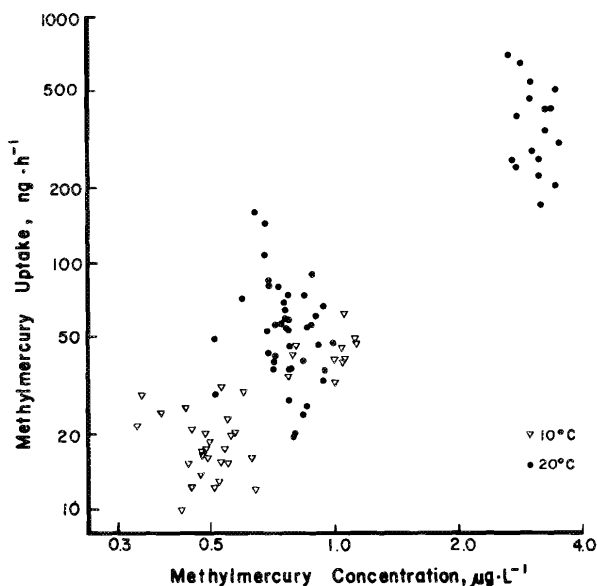


FIG. 4. Effect of methylmercury concentration on the rate of methylmercury uptake of rainbow trout at 10 and 20°C.

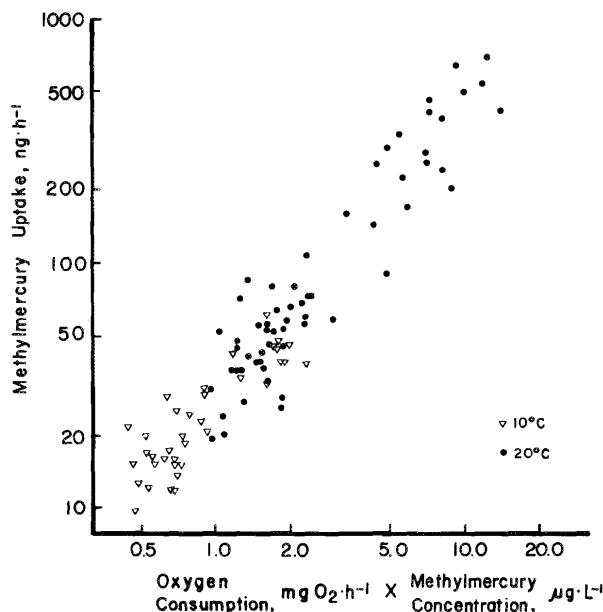


FIG. 5. Relationship of methylmercury uptake rate to the product of oxygen consumption and methylmercury concentration for rainbow trout at 10 and 20°C.

cantly different ($P < 0.05$). The relationship between $\log \Delta P$ and $\log (Q \times C_{pw})$ for individual fish at both temperatures is illustrated in Fig. 5. The regression equation describing this relationship is:

$$(6) \quad \log \Delta P = 1.445 + 1.134 \log (Q \times C_{pw})$$

$$(R^2 = 0.91; n = 98).$$

The efficiency of methylmercury uptake relative to oxygen was not significantly different ($P < 0.05$) at 10 and 20°C with an overall mean of ~ 0.25 (0.24 ± 0.01 (mean $\pm$ SE) at 10°C and 0.26 ± 0.01 at 20°C). This ratio was essentially independent of the rates of oxygen consumption and methylmercury uptake observed and the concentrations of oxygen and methylmercury used. Simple correlation coefficients between $E(p/o)$ and any of the above variables were less than 0.55.

Discussion

The concentrations ($< 5.0 \mu\text{g Hg} \cdot \text{L}^{-1}$) and durations of exposure to methylmercury we used were well below acutely toxic levels and did not affect rainbow trout as measured by effects on oxygen consumption. This is consistent with the observation that brook trout, *Salvelinus fontinalis*, exposed to methylmercury concentrations of less than $3 \mu\text{g Hg} \cdot \text{L}^{-1}$ showed no indication of stress as measured by cough frequency (Drummond et al. 1974). The 96-h LC50's of methylmercury to rainbow trout and brook trout are reported as 31 and $75 \mu\text{g Hg} \cdot \text{L}^{-1}$, respectively (Matida et al. 1971; McKim et al. 1976).

The rates of weight-specific oxygen consumption obtained at both 10 and 20°C are in agreement with metabolic rates previously reported for rainbow trout. The metabolic rates calculated from equations (1) and (2) for trout swimming at

6 BL/s, assuming a weight exponent of 1.0 (Rao 1968) (332 and $545 \text{ mg O}_2 \cdot \text{kg}^{-1} \cdot \text{h}^{-1}$ at 10 and 20°C, respectively), fall within the limits of active metabolism reported by Dickson and Kramer (1971) for two strains of rainbow trout (468–492 and $570\text{--}592 \text{ mg O}_2 \cdot \text{kg}^{-1} \cdot \text{h}^{-1}$ at 10 and 20°C, respectively). Using a weight exponent of 0.8 for standard metabolic rate (Rao 1968), the standard metabolic rates of 157 and $309 \text{ mg O}_2 \cdot \text{kg}^{-0.8} \cdot \text{h}^{-1}$ at 10 and 20°C for 6-g trout in this study are in agreement with standard metabolic rates of 143 and $278 \text{ mg O}_2 \cdot \text{kg}^{-0.8} \cdot \text{h}^{-1}$ reported for 100-g rainbow trout at 5 and 15°C (Rao 1968). The linear relationship between the log of weight-specific oxygen consumption and swimming speed is consistent with the pattern reported for this and other salmonids (Brett 1965; Rao 1968; Waiwood and Beamish 1978). The greater variability at 20°C may reflect the greater variability in size of fish tested at this temperature, or that fish at 20°C were stressed by temperature. Although the upper incipient lethal temperature for rainbow trout is 25.6°C, the optimum temperature for growth and yield is in the range of 17°C (Hokanson et al. 1977). In these growth experiments, at temperatures above optimum, mortality rates increased significantly (Hokanson et al. 1977), suggesting that at elevated sublethal temperatures the trout are stressed. A similar increase in the variability of regressions between weight-specific rate of oxygen consumption and swimming speed has been reported for rainbow trout stressed by sublethal concentrations of copper or rotenone (Waiwood and Beamish 1978; Skadsen et al. 1980).

We have demonstrated that the rate of uptake of waterborne methylmercury may be accurately described from oxygen consumption and methylmercury concentration. Our finding that regression equations relating methylmercury uptake, oxygen consumption, and methylmercury concentration at 10

and 20°C could be combined into a single equation substantiates the proposition of deFreitas and Hart (1975) that temperature affects methylmercury uptake by a direct effect on metabolic rate. It is probable that many previous reports of temperature effects on xenobiotic uptake (Murphy and Murphy 1971; Reinert et al. 1974) can be similarly explained. It is also apparent that in studies of xenobiotic accumulation, factors which affect metabolic rate such as handling and feeding may markedly affect the results obtained. As the toxicity of xenobiotics may relate to rates of uptake (Roberts et al. 1978), similar considerations may also apply to toxicity studies.

The relationship between concentration of methylmercury and rate of methylmercury uptake was linear for a duration of at least 24 d and was not affected by simultaneous exposure to dietary methylmercury (Phillips and Buhler 1978). Thus equation (6) should be valid for predicting long-term uptake of waterborne methylmercury.

The consistency of the efficiencies of uptake of methylmercury relative to oxygen at both temperatures suggests that this ratio (0.25) may be characteristic of the compound and experimental conditions used. The percent utilization of oxygen from water for rainbow trout swimming in well-aerated water was a constant value of 33%, at swimming speeds from less than 20% of critical to critical (Kiceniuk and Jones 1977). The percent utilization of methylmercury from water calculated using the value of 33% is 8.3%; a value consistent with a percent utilization of methylmercury of 7.1% reported by Boddington et al. (1979). Under conditions of oxygen stress imposed by low concentrations of dissolved oxygen, the percent utilization of oxygen from the water may decrease (Kerstens et al. 1979; McKim 1981). If the efficiency of uptake of pollutant relative to oxygen remains stable, the percent utilization of pollutant should decrease as well, which is consistent with the observations of McKim (1981) on the effects of low concentrations of dissolved oxygen on endrin uptake in brook trout.

Equations (3), (4), (5), and (6) do not require assumptions of the efficiencies of removal of methylmercury and oxygen from the water and thus represent a significant simplification over the expressions used by Norstrom et al. (1976) to describe methylmercury uptake from water. For our data, the replacement of the rate of oxygen consumption by the rate of oxygen consumption divided by mean oxygen concentration did not improve predictions of rate of methylmercury uptake, and the simpler model was selected.

A number of water quality factors such as pH, alkalinity, and the presence of selenium have been reported to affect methylmercury accumulation in wild fish (Scheider et al. 1979; Speyer 1980). These factors could affect accumulation either through effects on the levels or distribution of methylmercury (Miller and Akagi 1979) or through effects on uptake and retention by the fish. An analysis of the effects of these entities on the efficiency of uptake of methylmercury relative to oxygen may be of particular use in identifying the mechanisms by which they affect accumulation. Conversely, comparisons of the efficiency of pollutant uptake relative to oxygen of different pollutants under similar conditions may provide a sensitive standard index of the potential for pollutants to be accumulated from water.

As our data define the relationship between methylmercury uptake, oxygen consumption, and methylmercury concentration with precision, a measure of two of the above can provide an estimate of the third. Thus if the rate of oxygen consumption of a fish is known or can be estimated, a measure of the rate of uptake of waterborne methylmercury of the fish will provide an estimate of methylmercury concentration (see also Fagerstrom and Asell 1976). Further, as low concentrations of [^{203}Hg]methylmercury, such as were used in this study, did not affect the oxygen consumption of the fish, but are easily measured, a measure of the rate of uptake of [^{203}Hg]methylmercury by a fish under defined exposure concentrations gives an estimate of the fish's metabolic rate. This measurement may be particularly valuable in quantifying metabolic rate in situations not amenable to conventional techniques.

Acknowledgments

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